

Indian Institute of Technology Hyderabad

Deep Learning (AI2100/AI5100/EE6380): Test 1

Part A: Multiple Choice Questions and True/False (1 Mark Each)

- Q1.** How many hidden perceptrons are sufficient to represent any Boolean function of n inputs?
(A) $n + 1$ (B) 2^n (C) n^2 (D) 2^{n-1}
- Q2.** Gradient Descent minimizes a loss function by:
(A) Adding noise to weights
(B) Updating weights in the direction of the gradient
(C) Updating weights in the opposite direction of the gradient
(D) Randomly modifying weights
- Q3.** Which loss function is commonly used for regression problems?
(A) Cross-entropy (B) Mean squared error (C) Hinge loss (D) Logarithmic loss
- Q4.** Deep learning is preferred over shallow networks because it:
(A) Requires fewer parameters (B) Generalizes well (C) Enables hierarchical learning
(D) Both A and B (E) Both B and C
- Q5.** The Jacobian is used to compute:
(A) Gradient of scalar functions (B) Gradient of vector-valued functions
(C) Hessian matrices (D) None of the above
- Q6.** Which type of gradient descent updates weights after processing the entire training dataset?
(A) Batch Gradient Descent (B) Stochastic Gradient Descent
(C) Mini-batch Gradient Descent (D) None of the above
- Q7.** Gradient Descent always converges to the global minimum in non-convex loss surfaces. [T/F] Why?
- Q8.** Backpropagation calculates gradients using the forward pass only. [T/F] Why?
- Q9.** Mini-batch gradient descent updates weights after processing all examples. [T/F] Why?
- Q10.** A high learning rate can cause the loss function to oscillate around the minimum. [T/F] Why?

Part B: Subjective Questions

- Q1.** Write the perceptron learning algorithm. (2 Marks)
- Q2.** Consider the function $h(x, y) = e^{x+y^2}$, where $k = x + y^2$ and $h(x, y) = e^k$. Symbolically evaluate the gradients $\frac{\partial h}{\partial k}$, $\frac{\partial k}{\partial x}$, and $\frac{\partial h}{\partial y}$. (2 marks)
- Q3.** Neural Networks
- (a) Consider a deep neural network with a single input, a single output, and K hidden layers, each of which contains D hidden units. Find the total number of parameters. (2 Marks)

(b) Consider a three-layer neural network:

$$\begin{aligned} z_1 &= W_1 x + b_1, \\ z_2 &= W_2 z_1 + b_2, \\ z_3 &= W_3 z_2 + b_3, \end{aligned}$$

where: The input vector x has a shape of 8×1 . The neural network consists of three layers: the first hidden layer contains 10 neurons, the second hidden layer has 6 neurons, and the output layer consists of 2 neurons.

- i. Find the shapes of W_1 , b_1 , W_2 , b_2 , W_3 , and b_3 . (1 Marks)
- ii. Calculate the total number of parameters in the network. (1 Marks)

Q4. This problem explores computing derivatives on general acyclic computational graphs. Compute the derivative $\frac{\partial z}{\partial x}$, where $z = \sigma(y)$ by reverse-mode differentiation or forward-mode differentiation. (4 marks)

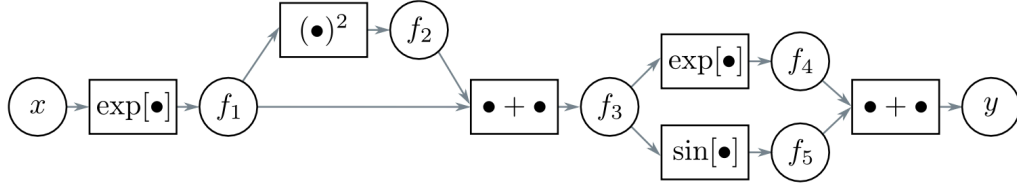


Figure 1: Computational graph